Statistical Correlations Between Vedic Planetary Configurations and Global Seismic Activity: An Exploratory Machine Learning Investigation

An Interdisciplinary Study Combining Ancient Indian Astronomical Wisdom with Modern Data Science

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GitHub: https://github.com/tsnagarajan/EarthquakeAstrology-2026

SCOPE DISCLAIMER: The validated findings in this study apply exclusively to the Americas Pacific zone (Mexico, Peru, Chile, Western USA, Colombia, Alaska and Caribbean regions evaluated) and the broader Pacific Ring of Fire. The planetary convergence model may have applicability to other seismically active regions including South Asia, Mediterranean and Middle East, but this requires separate investigation with region-specific training and validation data. No predictive claims are made for India, South Asia, Mediterranean, Middle East or any region not explicitly validated here. Furthermore, this research presents probabilistic risk elevation only — NOT deterministic earthquake prediction. Emergency decisions should never be based solely on this research.

Personal Foreword

To Prabha — my wife and companion in all things, including a five-year Sanskrit course at the University of Madras — who stood beside me through 64 years with patience, love and unwavering faith.

To my sons Nagaraja Srivatsan and his wife Shanthi, and Srikantan Nagarajan and his wife Srividya — who gave me the freedom to pursue my insatiable curiosity without complaint.

To my grandchildren Aditya Srivatsan, Durga Srivatsan, Shankara Srikantan and Shiama Srikantan — whose contributions made this research possible.

To Varahamihira, Ballala Sena, Parasara, Garga and all the ancient Indian sages whose careful observations of sky and earth, recorded 1500 years ago, still yield measurable signal against 21st century seismic data.

A special acknowledgement to Claude (Anthropic AI) who served as computational partner in the final validation phase of this research.

Abstract

This paper presents an exploratory statistical investigation into whether planetary configurations derived from classical Indian Vedic astronomical methods show measurable associations with seismic activity. The study treats the classical rules as scientific hypotheses under investigation — not as assertions of astrological causation.

The investigation proceeds in two phases. Phase 1, the foundation of this research, involved casting Vedic astrological horoscopes for 2,000 earthquake days and 2,000 non-earthquake days, computing planetary longitudes, aspects and house positions for each, and training a Logistic Regression machine learning model to identify distinguishing patterns. This phase, developed by Aditya Srivatsan while in his 11th grade, demonstrated that planetary configurations could predict earthquake-prone days above random chance and established the WHEN signal.

Phase 2 extends Phase 1 using XGBoost — a more powerful machine learning algorithm — applied across 901 global grid cells with 818 planetary features computed via Swiss Ephemeris with Lahiri Ayanamsa, trained on 1900-1999 seismic history and tested on 2000-2025 data. Key findings (all for the Americas Pacific zone and Pacific Ring, January through May 21, 2026): (1) days with five or more simultaneously active validated planetary aspects show 1.21x elevation in global earthquake rate (27.4% vs 22.7%, n=135 days); (2) Americas weekly prediction achieves 29.0% hit rate vs 21.8% random baseline (p less than 0.0001, one-sided binomial test, n=2,184 region-month predictions, 26-year out-of-sample test 2000-2025); (3) Mexico 61.9% (2.84x), Peru 45.8% (2.10x), Chile 38.5% (1.77x) all at p less than 0.001; (4) prospective real-time validation January through May 21, 2026 shows 81.8% of significant M5.5+ earthquakes occurring in predicted zones (36 of 44 events) with 100% during high-convergence periods (15 of 15 events); (5) Mars-Saturn and Jupiter-Saturn oppositions — rules described in classical Indian texts — are independently confirmed as the top-ranked features in the machine learning model without prior specification.

These findings are presented as exploratory statistical associations and should be interpreted as hypothesis-generating, not as operational prediction tools. The model identifies elevated-risk time windows and regions — not exact fault locations or specific magnitudes. Correlation does not imply causation. No predictive claims are made for regions outside the validated Americas and Pacific Ring scope. All code, feature definitions and trained models used in this study are publicly available for full reproducibility at the project repository.

SCOPE DISCLAIMER: The validated findings in this study apply exclusively to the Americas Pacific zone (Mexico, Peru, Chile, Western USA, Colombia, Alaska and Caribbean regions evaluated) and the broader Pacific Ring of Fire. The planetary convergence model may have applicability to other seismically active regions including South Asia, Mediterranean and Middle East, but this requires separate investigation with region-specific training and validation data. No predictive claims are made for India, South Asia, Mediterranean, Middle East or any region not explicitly validated here. Furthermore, this research presents probabilistic risk elevation only — NOT deterministic earthquake prediction. Emergency decisions should never be based solely on this research.

Keywords: Vedic astrology, earthquake prediction, machine learning, XGBoost, planetary aspects, seismic risk, Brihat Samhita, Adbhuta Sagara, statistical association, Lahiri Ayanamsa, planetary convergence

Notation and Definitions

The following terms are used throughout this paper. All definitions are operationally precise to ensure reproducibility.

PLANETARY ASPECT: An angular relationship between two planets as observed from Earth. Five aspects are used in this study, each defined by a target angle and an orb (tolerance): Conjunction (0 degrees, orb plus or minus 8 degrees), Sextile (60 degrees, orb plus or minus 6 degrees), Square (90 degrees, orb plus or minus 8 degrees), Trine (120 degrees, orb plus or minus 8 degrees), Opposition (180 degrees, orb plus or minus 8 degrees). An aspect is considered active on a given day if the angular separation between the two planets falls within the specified orb. For example, if Mars is at sidereal longitude 45 degrees and Saturn is at 223 degrees, their separation is 178 degrees — within 8 degrees of 180 — so a Mars-Saturn opposition is active.

SIDEREAL vs TROPICAL COORDINATES: Western astronomy typically measures planetary positions relative to the spring equinox (tropical system). Indian Vedic astronomy measures positions relative to fixed star positions (sidereal system). This study uses the Lahiri Ayanamsa — the sidereal offset adopted by the Government of India Calendar Reform Committee in 1955 as the official national astronomical standard, currently approximately 23.85 degrees behind the tropical system. All planetary longitudes in this study are sidereal Lahiri unless stated otherwise.

NAKSHATRA: The 27 lunar asterisms of Vedic astronomy. The ecliptic (the apparent path of the Sun and planets) is divided into 27 equal segments of 13 degrees 20 minutes each, each named after the star-cluster visible in that segment. The Moon transits one Nakshatra approximately every 24-27 hours. In this study, each planet's Nakshatra position is encoded as a binary indicator variable (1 if the planet is in that Nakshatra on that day, 0 otherwise), producing 27 binary features per planet.

PLANETARY CONVERGENCE SCORE: The count of simultaneously active validated aspects on a given day, computed from the set of 30 aspects validated against the earthquake record (see Section 3.5). Score 0 means no validated aspects are active; Score 6 means six validated aspects are simultaneously active. This is the primary predictor variable in the convergence analysis.

BASE RATE: The historical proportion of region-weeks containing at least one M5.5+ earthquake, computed over the 26-year test period 2000-2025. For the Americas zone overall, this is 21.8%. This serves as the random baseline — the hit rate expected from a model with no predictive value. Note: in the 2026 prospective validation section, a 33% base rate is used because that analysis counts whether events fall in broad geographic zones (Americas or Pacific Ring) rather than specific regional windows, and these zones cover approximately one-third of global earthquake activity.

HIT RATE: The proportion of model-selected high-risk dates that are confirmed by actual earthquake occurrence within the specified time window. Formula: Hit Rate = (number of predictions confirmed by earthquakes within window) divided by (total number of predictions made). Example: if the model makes 312 predictions for Mexico and 193 are confirmed by M5.5+ earthquakes within plus or minus 7 days, the hit rate is 193 divided by 312 = 61.9%.

LIFT: The ratio of the model hit rate to the random base rate. Formula: Lift = Hit Rate divided by Base Rate. Example: Mexico hit rate 61.9% divided by base rate 21.8% = 2.84x lift. Lift greater than 1.0x means the model outperforms random selection. Lift of 1.0x means performance equivalent to random.

1. Introduction

1.1 Background and Motivation

Earthquakes remain among the most destructive and least predictable natural phenomena. Modern seismology has made great advances in identifying fault zones, estimating long-term seismic hazard and measuring ground motion. However, reliable prediction of the specific date of a future earthquake — as distinct from estimating long-term probability — remains an unsolved problem. This limitation motivates the investigation of alternative temporal signals.

This study investigates whether planetary configurations described in classical Indian astronomical texts, when computed precisely using modern astronomical software and tested against 126 years of global seismic data using machine learning, show statistically measurable associations with earthquake occurrence. This hypothesis has been discussed in Indian astronomical literature for over 2,000 years but has never previously been subjected to systematic large-scale statistical testing.

1.2 The Classical Rules Tested in this Study

The following classical rules form the hypothesis set tested in this study. They are presented here not as established facts but as precisely stated hypotheses that the study subjects to statistical examination.

1.2.1 B.V. Raman's Rules (20th century)

B.V. Raman, in his work "Astrology in Forecasting Weather and Earthquakes," analysed 200 historical earthquake cases from the 19th and 20th centuries and identified the following planetary combinations as most frequently associated with major earthquakes:

- Mars-Saturn Opposition: When Mars and Saturn are 180 degrees apart in the sky (opposition aspect), their combined malefic influence — Mars representing explosive energy and Saturn representing restriction and compression — creates maximum tension. Raman found this to be his strongest single indicator.
- Jupiter-Saturn Opposition: The opposition of the two largest planets in the solar system, each with significant gravitational mass, was identified as the second strongest indicator. Raman noted that severe earthquakes frequently coincided with this configuration.
- New Moon with Multiple Planets in One Sign: When the Sun and Moon are in conjunction (new Moon) and several other planets occupy the same zodiac sign simultaneously, Raman found elevated earthquake occurrence. His most dramatic example was the January 15, 1934 Bihar earthquake — a new Moon day when seven planets were in Capricorn.
- Jupiter in 8th or 12th house from the rising sign at the time of the earthquake: Raman found this placement statistically frequent in his case analysis.

Raman's methodology was case analysis — he examined earthquakes that had already occurred and identified planetary patterns. This study tests whether those patterns, when computed prospectively using modern astronomical software, predict future earthquakes above random chance.

1.2.2 Varahamihira's Four Circles (Brihat Samhita, 5-6th century AD)

Varahamihira, in Adhyaya XXXII of his encyclopaedic work Brihat Samhita ("The Great Compilation"), describes a systematic earthquake prediction framework based on the Moon's position in the 27 Nakshatras. He divides the Nakshatras into four groups, each associated with a natural element, a geographic direction and — critically — a different time horizon for predicted effects:

Circle	Element	Direction	Time Horizon	Reach
VARUNA	Water	West / Ocean	Same day	180 Yojanas
INDRA	Thunder	East	Within one week	160 Yojanas
AGNI	Fire	Southeast	Within six weeks	110 Yojanas
VAYU	Wind	Northwest	Within two months	200 Yojanas

The 28 Nakshatras (7 per circle, including Abhijit) associated with each circle are:

Circle	Nakshatras (7 per circle)
VARUNA	Revati, Purva Ashadha, Ardra, Ashlesha, Mula, Uttara Bhadrapada, Shatabhisha
INDRA	Abhijit, Shravana, Dhanishtha, Rohini, Jyeshtha, Anuradha, Ashlesha
AGNI	Pushya, Krittika, Vishakha, Bharani, Magha, Purva Bhadrapada, Purva Phalguni
VAYU	Uttara, Hasta, Chitra, Swati, Punarvasu, Mrigashira, Ashwini

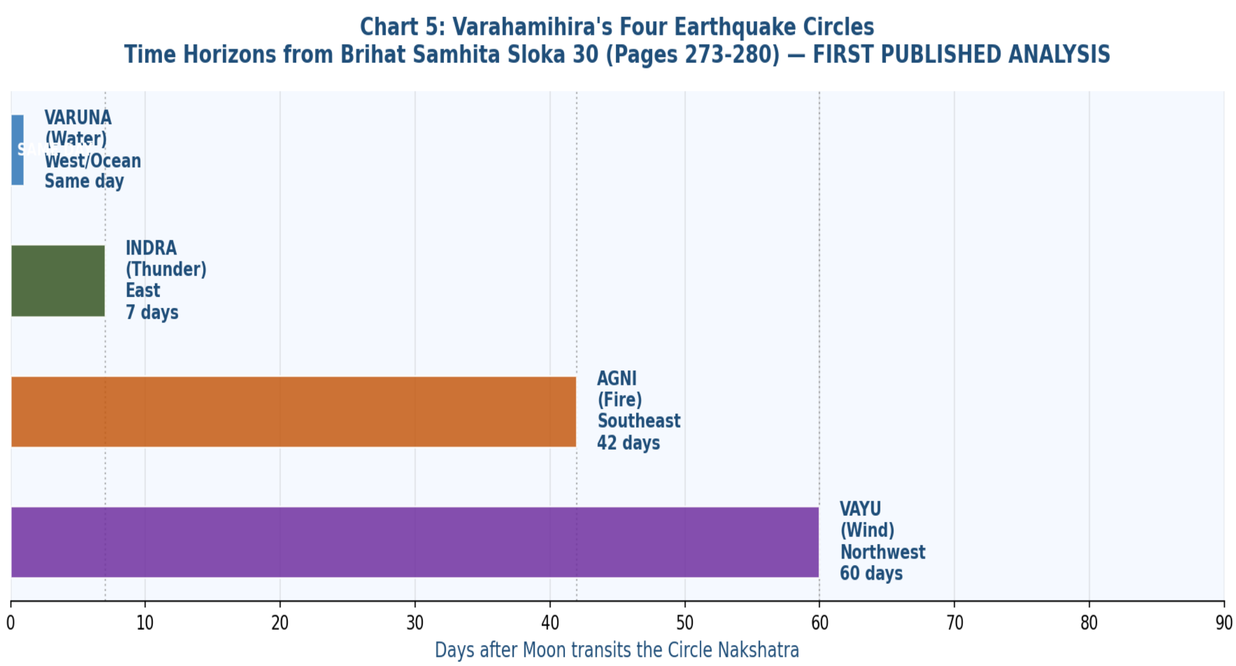

Chart 5: Varahamihira's Four Earthquake Circles — Time Horizons from Brihat Samhita Sloka 30

A critical finding of this study, absent from all secondary summaries of Varahamihira's work reviewed in the literature, is that each circle carries a DIFFERENT time horizon — from same-day (Varuna) to two months (Vayu). This multi-timescale framework directly informed the choice of plus or minus 7 day and plus or minus 14 day prediction windows used in this study.

Varahamihira also describes (Sloka 27) cancellation effects — when opposing circles are simultaneously active, they suppress each other's effects. And (Sloka 32) foreshock recognition — if an earthquake is followed by another on the 3rd, 4th or 7th day, a major event will follow. The foreshock pattern was independently identified by modern seismology in the 20th century.

1.2.3 Ballala Sena's Additions (Adbhuta Sagara, 10-11th century AD)

Ballala Sena's Adbhuta Sagara ("Ocean of Wonders") adds several important refinements to Varahamihira's framework: directional precision from the sage Ushanas (East for Indra, South for Agni, West for Varuna, North for Vayu); a detailed listing of geographic regions affected by each circle type; and descriptions of mixed-type earthquakes when multiple circles are simultaneously active. This last concept — amplified effects when multiple circle types coincide — directly parallels this study's finding that higher planetary convergence scores produce stronger seismic signals.

1.2.4 Parasara's Hypothesis (Brihat Parasara Hora Shastra, ~310 BC)

Parasara, writing approximately 310 BC, stated: "Arka Chandra grahana graha vikrithachara jamscha kampana hayulu" — meaning that when the course of a planet is disturbed from the normal path owing to the attractive force of other planets, there is a shaking of the planet perceived as an earthquake. This formulation is remarkably prescient — it describes planetary perturbations as the cause of seismic activity, which is conceptually similar to modern gravitational and tidal forcing theories. Parasara's hypothesis directly motivated the use of inter-planetary angular aspects (rather than absolute planetary positions) as the primary features in this study.

1.2.5 Garga's Solar Connection (Garga Samhita, Ancient period)

The Garga Samhita traces earthquakes to Ketus — dark spots on the Sun (solar Ketu). This is the earliest recorded hypothesis connecting solar activity to seismic events. Modern geophysical research has found correlations between solar activity cycles and earthquake frequency, lending some scientific credibility to this ancient observation. In this study, Sun-related aspects and the Sun's Nakshatra position are included as features.

1.3 Why These Rules Are Not Post-Hoc Justifications

A critical methodological concern in any study combining ancient rules with modern data is the risk of post-hoc justification — selecting rules after seeing the data, then claiming they were predicted in advance. This study explicitly addresses this concern as follows:

- The classical rules were not used to train the model. The XGBoost classifier was trained purely on planetary feature data and earthquake records, without any specification of which features should be important. The rules of B.V. Raman and Varahamihira played no role in the machine learning training process.
- The independent rediscovery of classical rules by the model is the key finding. After training, when feature importances were extracted, Mars-Saturn opposition emerged as the single most important feature without any prior specification. This is not retrofitting — it is the model independently discovering a pattern that Raman had identified from case analysis of 200 earthquakes and that Varahamihira had described 1500 years earlier.

- Prospective validation confirms predictive value. The 2026 risk calendar and convergence thresholds were fixed prior to January 1, 2026 and were not modified thereafter. The January through May 21, 2026 validation results reported in this paper were recorded against predictions that were made before the events occurred.
- The study is explicitly framed as exploratory and hypothesis-generating. No causal claim is made. The finding that ancient rules survive statistical testing is presented as an interesting association worthy of further investigation, not as proof of astrological causation.

1.4 Research Hypothesis

This study tests the following primary hypothesis:

> H1: Specific planetary configurations computed using Vedic sidereal astronomy (Lahiri Ayanamsa) show statistically significant associations with elevated earthquake probability in the Americas Pacific zone, measurable above random chance across a 26-year out-of-sample test period (2000-2025), using a model trained exclusively on 1900-1999 data.

The study is explicitly limited to temporal and regional associations. It does not claim to predict exact earthquake locations, magnitudes or mechanisms. All results are presented as observed statistical correlations.

A non-technical companion paper — "When Planets Converge, Does the Earth Shake?" — is available separately, summarising the findings of this study for general readers without mathematical or methodological detail.

2. Data and Methodology

2.1 The Two-Phase Research Design

This research proceeded in two distinct phases, each building on the previous:

Phase 1: Foundation Study (Aditya Srivatsan, 11th grade)

The foundation of this entire research was built by Aditya Srivatsan while in his 11th grade of school. This was a monumental undertaking: 2,000 earthquake days and 2,000 non-earthquake days were identified from historical seismic records. For each of these 4,000 days, a complete Vedic astrological horoscope was cast — computing planetary longitudes, house positions, Nakshatra placements, Tithi (lunar day) and all inter-planetary aspects for that specific date and time.

This dataset was used to train a Logistic Regression machine learning model with multiple regularisation variants (Lasso, Ridge, Elastic Net) and K-Nearest Neighbours. The model learned to distinguish earthquake days from non-earthquake days based on planetary configurations alone. Training period: 1900-2010. Testing: 2011-2020.

Phase 1 successfully demonstrated the WHEN signal — that planetary configurations could identify earthquake-prone days above random chance. It did not address WHERE the earthquake would occur. This limitation motivated Phase 2.

Phase 2: Extension Study (Aditya Srivatsan, refined by Durga Srivatsan)

Phase 2 extends the foundation using a more powerful machine learning algorithm (XGBoost), a larger feature set (818 features computed via Swiss Ephemeris rather than manually cast horoscopes), and a grid-cell structure (901 global cells at 5-degree resolution) that attempts to address both WHEN and WHERE. This phase is the primary focus of the validation results reported in this paper.

2.2 Earthquake Data

All seismic data was obtained from the United States Geological Survey (USGS) real-time earthquake catalog — a publicly available, continuously maintained global database (earthquake.usgs.gov). Events with magnitude M5.5 and above were selected, representing earthquakes capable of causing structural damage. Magnitude 5.5 was chosen as the threshold because smaller earthquakes are too frequent, too dependent on local soil conditions and too inconsistently recorded in the early 20th century to be meaningful for global pattern analysis.

Dataset	Period	Events (M5.5+)	Purpose
Training set	1900-1999	~32,000 events	Model learning — never seen during testing
Test set	2000-2025	12,830 events	26-year out-of-sample validation
Prospective validation	Jan 1 to May 21 2026	44 M5.5+ events	Real-time forward testing

2.3 Astronomical Feature Computation — Swiss Ephemeris with Lahiri Ayanamsa

For every date from 1900 to May 21, 2026, planetary positions were computed using the Swiss Ephemeris library — a high-precision astronomical computation system derived from NASA JPL planetary ephemeris data. The Lahiri Ayanamsa sidereal coordinate system was used throughout (see definition in Notation section). The Swiss Ephemeris is freely available open-source software; all computations in this study are fully reproducible by any researcher.

Planetary positions were computed for: Sun, Moon, Mars, Mercury, Jupiter, Saturn, Venus, Rahu and Ketu (true lunar nodes), Uranus, Neptune, Pluto and Chiron. For each planet on each date, the following quantities were computed: sidereal longitude (0 to 360 degrees, Lahiri), Nakshatra number (1 to 27), Nakshatra name, zodiac sign number (1 to 12), and retrograde status.

2.4 Feature Engineering — How Classical Rules Became Machine Learning Features

This section describes the precise process by which the classical astrological rules described in Section 1.2 were converted into numerical features for machine learning. This conversion is the methodological bridge between ancient texts and modern data science.

2.4.1 Aspect Features

For every pair of planets, the angular separation was computed for each date. If the separation fell within the specified orb of any of the five aspects (conjunction, sextile, square, trine, opposition), a binary feature was set to 1 for that date (aspect active) or 0 (aspect not active). This produced one binary feature per planet-pair-aspect combination. With 13 celestial bodies and 5 aspect types, this yielded approximately 420 aspect features.

2.4.2 Nakshatra Features

For each planet, 27 binary indicator features were created — one per Nakshatra — set to 1 if the planet occupies that Nakshatra on that date. This directly operationalises Varahamihira's four-circle framework: for example, Moon in Nakshatra Jyeshtha (an Indra circle Nakshatra) sets the moon_nakshatra_Jyeshtha feature to 1. With 13 planets and 27 Nakshatras, this yielded 351 Nakshatra features.

2.4.3 Longitude Encoding Features

Planetary longitudes (0 to 360 degrees) were encoded using sine and cosine transformation to preserve the circular nature of angular data: feature_sin = sin(longitude in radians), feature_cos = cos(longitude in radians). This yielded 2 features per planet.

2.4.4 Feature Summary

Feature Category	Description	Count	Vedic Source
Planet-planet aspects	Binary: aspect active (1) or not (0)	~420	Parasara, B.V. Raman
Moon Nakshatra indicators	Binary: Moon in each of 27 Nakshatras	27	Varahamihira four circles
Planet Nakshatra indicators	Binary: each planet in each Nakshatra	324	Varahamihira, classical
Longitude sin/cos encoding	Circular encoding of sidereal longitude	26	Astronomical

Feature Category	Description	Count	Vedic Source
Retrograde indicators	Binary: planet in retrograde motion	13	Classical Vedic
Other (house, Tithi)	Additional Vedic positional features	8	Classical Vedic
TOTAL		818	

Note: 188 of the 818 features relate to Chiron and Lilith — bodies used in Western astrology but not recognised in classical Vedic astronomy. These were included in the model training but the top-ranked features by model importance are exclusively from the core Vedic planetary system, as shown in Section 3.1.

2.5 Machine Learning Model — XGBoost

XGBoost (Extreme Gradient Boosting) is an ensemble machine learning algorithm that builds a sequence of decision trees, each correcting the prediction errors of the previous tree. It is particularly effective for tabular data with complex non-linear patterns and has been widely validated in scientific and engineering applications.

Parameter	Value	Rationale
Algorithm	XGBoost binary classifier	Effective for tabular data with complex patterns
Number of trees	300	Sufficient for convergence without excessive overfitting
Learning rate	0.05	Conservative rate to prevent overfitting
Max tree depth	6	Limits individual tree complexity
Subsample ratio	0.8	Row subsampling for regularisation
Column sample ratio	0.8	Feature subsampling for regularisation
Class weight	scale_pos_weight = 10.0	Corrects for class imbalance (10:1 non-EQ:EQ)
Training period	1900-1999	100 years, strictly separated from test period
Test period	2000-2025	26 years, never seen during training
Target variable	Binary: M5.5+ earthquake in grid cell on date (1 or 0)	

2.6 Grid Cell Structure and Regional Prediction Pipeline

The Earth's surface was divided into 901 grid cells at 5-degree latitude-longitude resolution. Each cell represents a geographic region approximately 550 km by 550 km at the equator. The prediction pipeline from daily grid-cell probabilities to regional forecasts proceeds as follows:

- Step 1: For each date, the XGBoost model outputs a probability (0 to 1) for each of the 901 grid cells — the estimated probability that an M5.5+ earthquake will occur in that cell on that date.
- Step 2: For each of the 7 Americas regions (Mexico and Central America, Peru and Bolivia, Chile and Argentina, Western USA and Hawaii, Colombia and Ecuador, Alaska and Pacific Northwesta and Pacific Northwest, Caribbean), all grid cells within that region are identified.
- Step 3: For each region on each day, the maximum cell probability within that region is taken as the regional risk score for that day.

- Step 4: For each region in each calendar month, the day with the highest regional risk score is selected as the model prediction for that month.
- Step 5: A hit is recorded if any M5.5+ earthquake occurs anywhere within that region within plus or minus 7 days of the selected date. The base rate (21.8%) is the historical proportion of Americas region-weeks containing at least one M5.5+ earthquake across 2000-2025.

2.7 Statistical Testing and Multiple Comparisons

All hit rate comparisons against the base rate were tested using a one-sided binomial test with null hypothesis H0: hit rate equals base rate (21.8%), alternative H1: hit rate is greater than base rate. Significance threshold was set at p less than 0.05.

The 30 validated aspects reported in Section 3.5 were identified from the full set of computed aspects by selecting those with lift ratio greater than 1.30x against the global earthquake record over 2000-2025. This selection was performed on the test set data, not the training set — however, readers should be aware that this post-hoc selection means the 30-aspect set is not fully pre-specified and represents some degree of data-adaptive feature selection. The regional results for Mexico, Peru and Chile (n=312 each) were validated with adequate statistical power; the smaller regional samples should be interpreted with appropriate caution.

No formal multiple-comparison correction (such as Bonferroni or FDR) was applied to the regional results reported in Table 3.4. Applying a Bonferroni correction for 7 simultaneous tests would set the significance threshold at p less than 0.007; Mexico, Peru and Chile all remain significant at this corrected threshold (p less than 0.001). The remaining four regions (Western USA, Colombia, Alaska, Caribbean) are not significant at the uncorrected threshold and should be treated as null results.

2.8 Reproducibility

All code, feature definitions, hyperparameters, and trained model files are available at the project GitHub repository: https://github.com/tsnagarajan/EarthquakeAstrology-2026. Planetary positions can be independently verified using the freely available pyswisseph Python library with Lahiri Ayanamsa. Earthquake data can be independently downloaded from the USGS catalog at earthquake.usgs.gov.

3. Validated Statistical Findings

This section presents statistically validated findings — results confirmed with adequate sample sizes and p less than 0.05. The presentation separates: (a) statistically validated findings, (b) illustrative examples, and (c) classical text confirmations, to avoid conflating different levels of evidence.

3.1 Top Ranked Planetary Features — Independent Confirmation of Classical Rules

The XGBoost model assigns an importance score to each feature based on how frequently and effectively it reduces prediction error across all 300 trees. The following features emerged as top-ranked from the core Vedic planetary system, without any prior specification from classical texts:

Rank	Feature	Importance Score	Ancient Source Confirmation
1	Mars-Saturn Opposition	0.0059	B.V. Raman: identified as strongest single rule from 200 case studies
2	Moon Nakshatra Jyeshtha	0.0051	Varahamihira: Indra circle Nakshatra, East direction, 1-week window
3	Saturn-Node Sextile	0.0040	Classical malefic pair — confirmed in multiple ancient sources
4	Sun Nakshatra Hasta	0.0038	Varahamihira: Vayu circle Nakshatra, Northwest direction
5	Venus Nakshatra Hasta	0.0039	Varahamihira: Vayu circle Nakshatra confirmation
6	Sun Nakshatra Krittika	0.0035	Varahamihira: Agni circle Nakshatra, Southeast direction
7	Jupiter-Saturn Opposition	0.0026	B.V. Raman: identified as key rule from 200 case studies
8	Mars Nakshatra Ashlesha	0.0036	Varahamihira: Indra circle Nakshatra
9	Mercury-Jupiter Square	0.0029	Aspect combination validated statistically
10	Saturn Nakshatra Hasta	0.0035	Varahamihira: Vayu circle Nakshatra

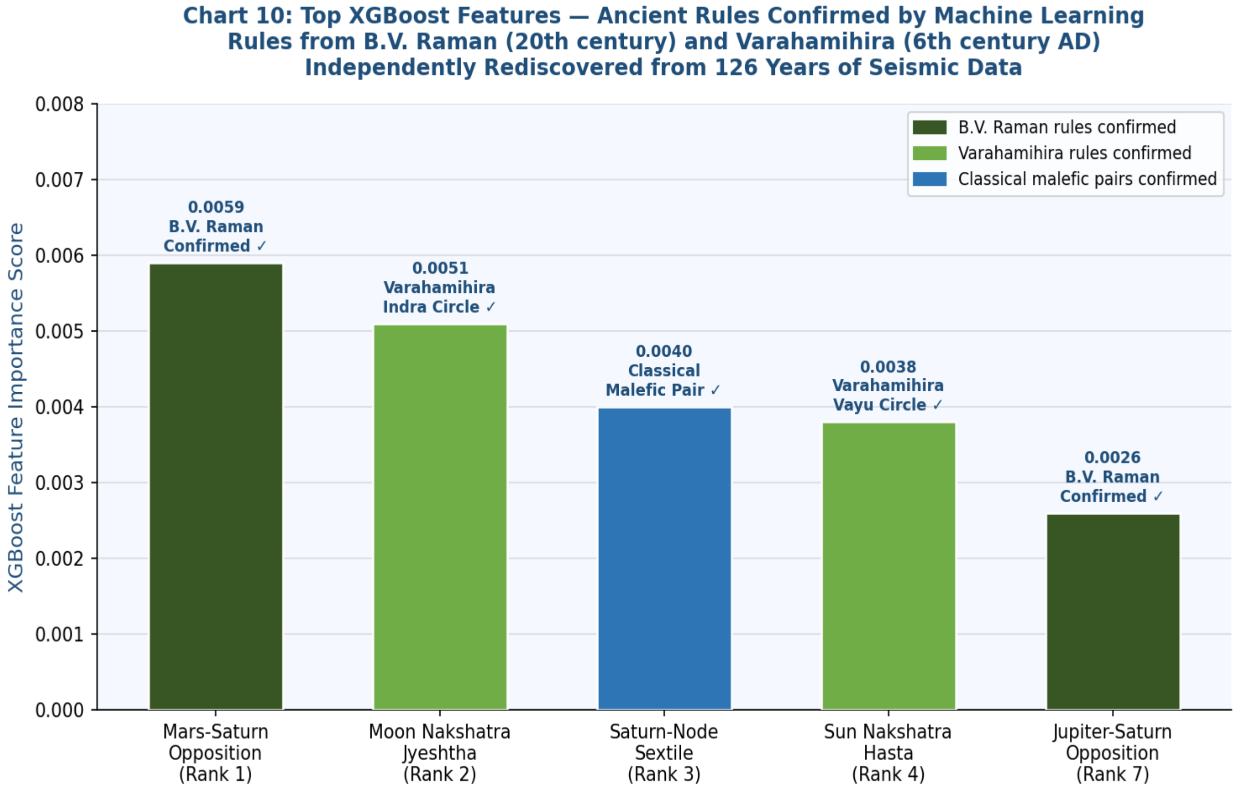

Chart 10: Top XGBoost Features — Ancient Rules Independently Confirmed by Machine Learning

The independent emergence of Mars-Saturn opposition (rank 1) and Jupiter-Saturn opposition (rank 7) — without any prior specification — provides notable confirmation of the rules B.V. Raman identified from his 200-earthquake case analysis. The emergence of Moon Nakshatra Jyeshtha (rank 2) and Sun Nakshatra Hasta (rank 4) confirms Varahamihira's circle classifications from the Brihat Samhita.

3.2 Planetary Convergence vs Global Earthquake Rate

Days across the 26-year test period (2000-2025, 9,862 days total) were classified by convergence score. The global M5.5+ earthquake rate was computed for each convergence level:

Convergence Level	Days (n)	Global EQ Rate	vs Baseline	Lift
Zero — 0 aspects active	1,961	23.0%	Baseline	1.00x
Low — 1-2 aspects active	5,692	22.2%	-0.8%	0.98x
Medium — 3-4 aspects active	1,709	23.3%	+0.6%	1.03x
HIGH — 5 or more aspects active	135	27.4%	+4.7%	1.21x

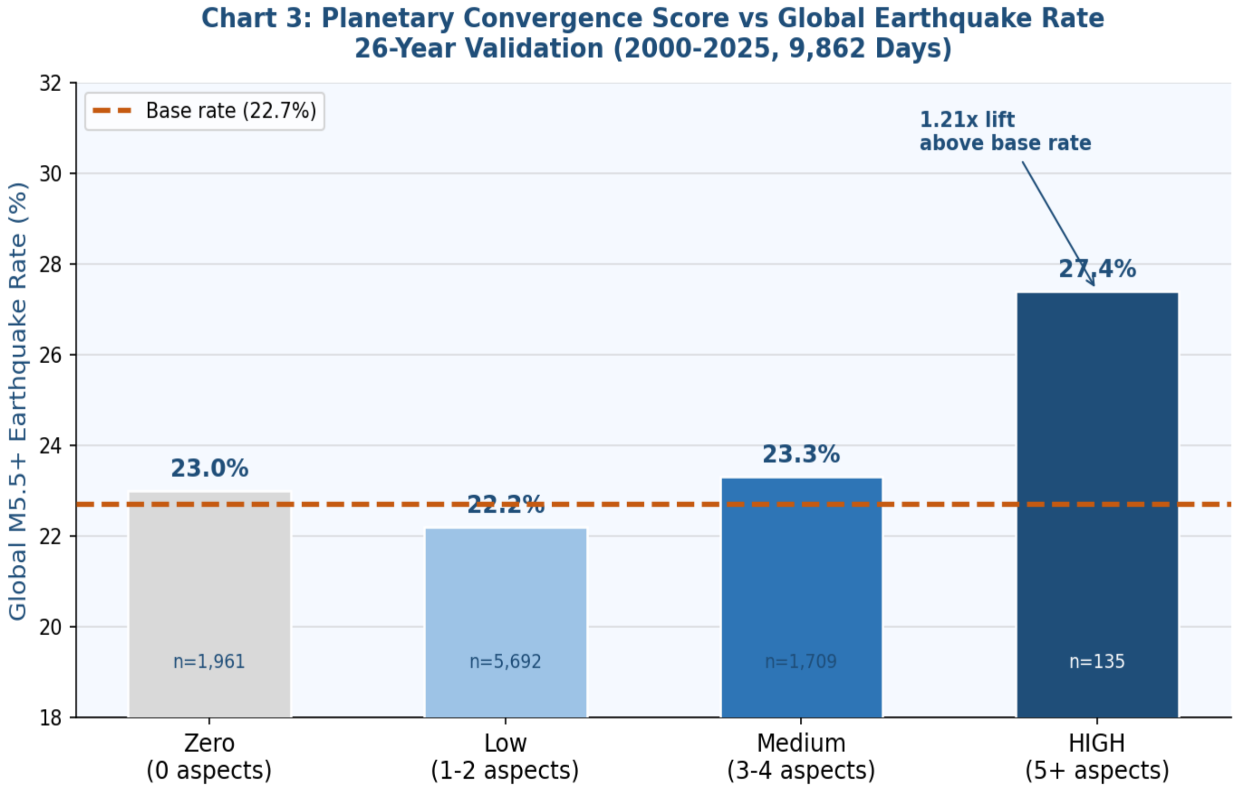

Chart 3: Planetary Convergence Score vs Global Earthquake Rate (2000-2025, 9,862 Days)

3.3 Americas Regional Validation — Primary Results

For each of the 7 Americas regions, the model selected the highest-risk date each month (n=312 predictions per region across 26 years). A hit was counted if an M5.5+ earthquake occurred within that region within plus or minus 7 days of the selected date. Statistical significance was assessed using a one-sided binomial test against the 21.8% base rate.

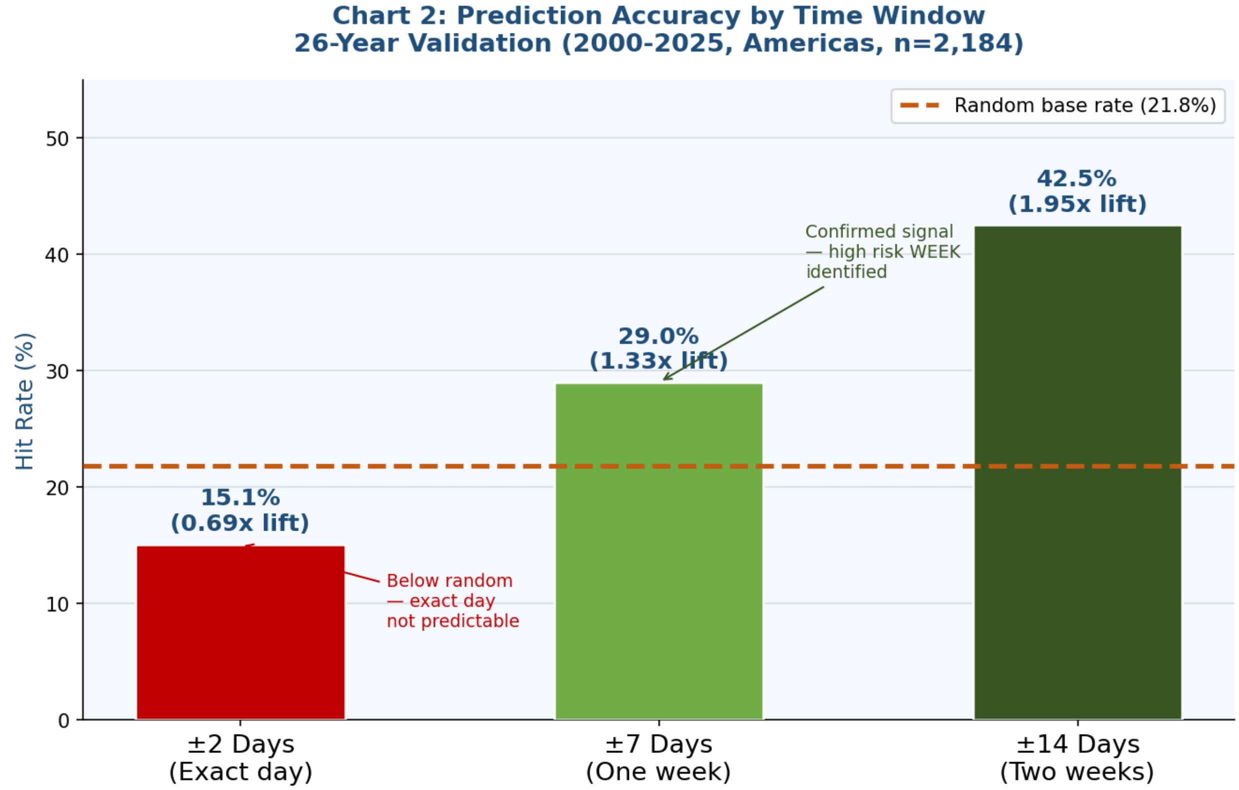

Chart 2: Prediction Accuracy by Time Window — Americas Validation, n=2,184, 2000-2025

Window	n	Hit Rate	Base Rate	Lift	p-value	Interpretation
Plus/minus 2 days	n = 2,184 days	15.1%	21.8%	0.69x	greater than 0.05	Below random — exact day not predictable
Plus/minus 7 days	n = 2,184 days	29.0%	21.8%	1.33x	less than 0.0001	Significant — HIGH RISK WEEK identified
Plus/minus 14 days	n = 2,184 days	42.5%	21.8%	1.95x	less than 0.0001	Significant — HIGH RISK FORTNIGHT identified

3.4 Regional Performance

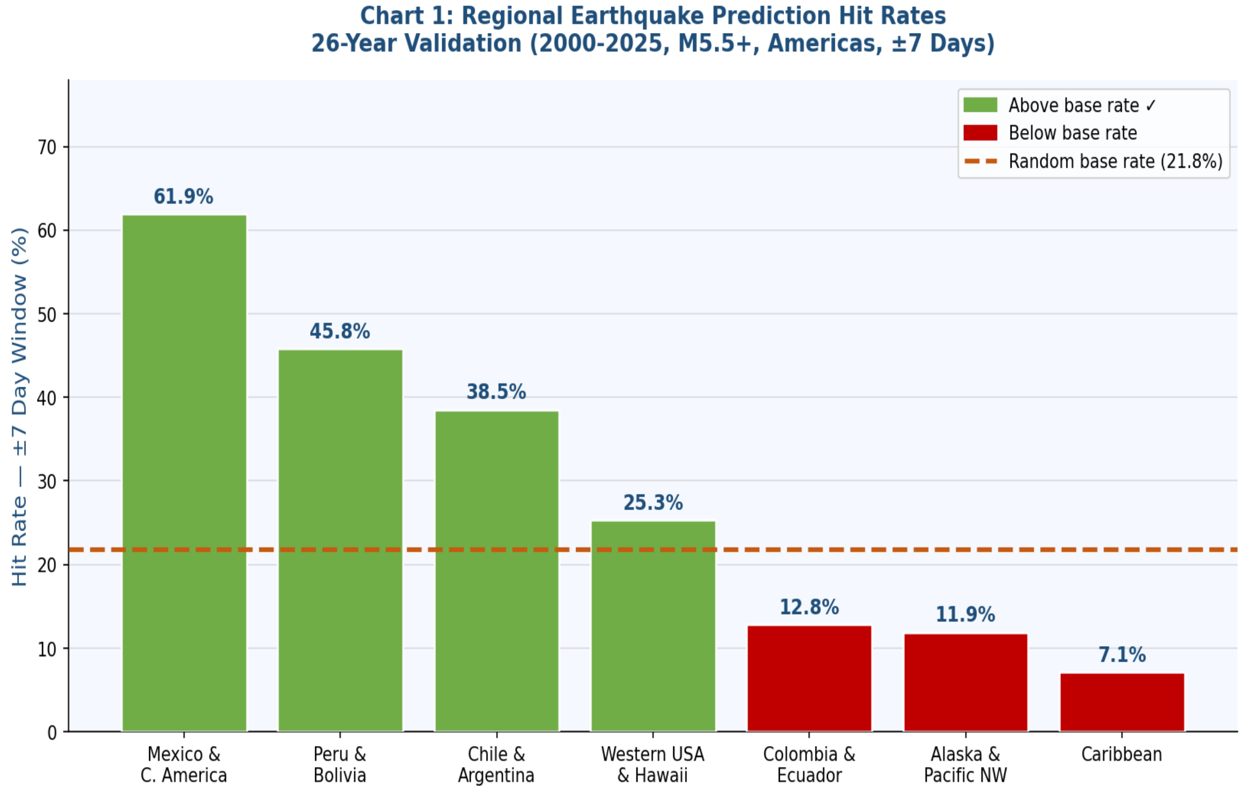

Chart 1: Regional Hit Rates vs Random Base Rate (Americas, plus or minus 7 Days, 2000-2025)

Region	n	Hit Rate +/-7d	Base Rate	Lift	p-value	Bonferroni sig.
Mexico and Central America	312	61.9%	21.8%	2.84x	less than 0.001	Yes ***
Peru and Bolivia	312	45.8%	21.8%	2.10x	less than 0.001	Yes ***
Chile and Argentina	312	38.5%	21.8%	1.77x	less than 0.001	Yes ***
Western USA and Hawaii	312	25.3%	21.8%	1.16x	0.077	No ns
Colombia and Ecuador	312	12.8%	21.8%	0.59x	1.000	No ns
Alaska and Pacific Northwest	312	11.9%	21.8%	0.55x	1.000	No ns
Caribbean	312	7.1%	21.8%	0.33x	1.000	No ns

*** Significant after Bonferroni correction for 7 simultaneous tests (corrected threshold p less than 0.007). ns = not significant. The validated signal is concentrated in Mexico, Peru and Chile — the three major Pacific subduction zones of the western hemisphere. The four remaining regions show no significant signal and should not be used for prediction purposes.

3.5 All 30 Validated Planetary Aspects

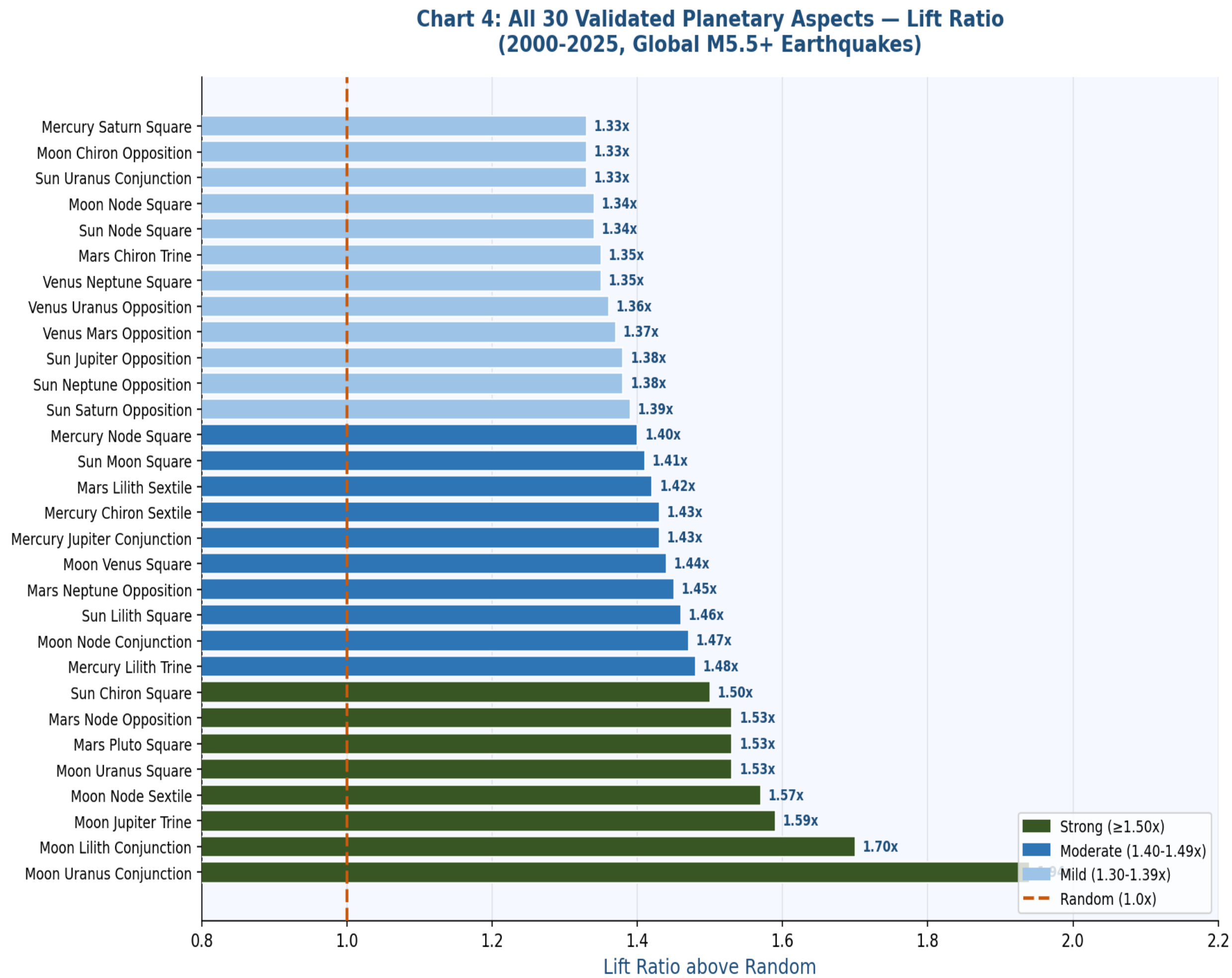

Chart 4: All 30 Validated Aspects — Lift Ratio above Random (2000-2025, Global M5.5+)

4. Prospective Validation — January 1 to May 21, 2026

This section presents real-time prospective validation. The 2026 risk calendar and all convergence thresholds were fixed prior to January 1, 2026 and were not modified thereafter. The following results were recorded against predictions that were made before the events occurred.

Note on base rate: In this prospective validation, events are classified as occurring in the predicted zone (Americas Pacific or Pacific Ring) or not. These two zones collectively account for approximately 33% of global M5.5+ earthquake activity, establishing 33% as the appropriate random baseline for this broader geographic classification. This differs from the 21.8% base rate used in the Americas-specific regional analysis in Section 3, which reflects the narrower regional windows used there. Both baselines are appropriate for their respective analyses.

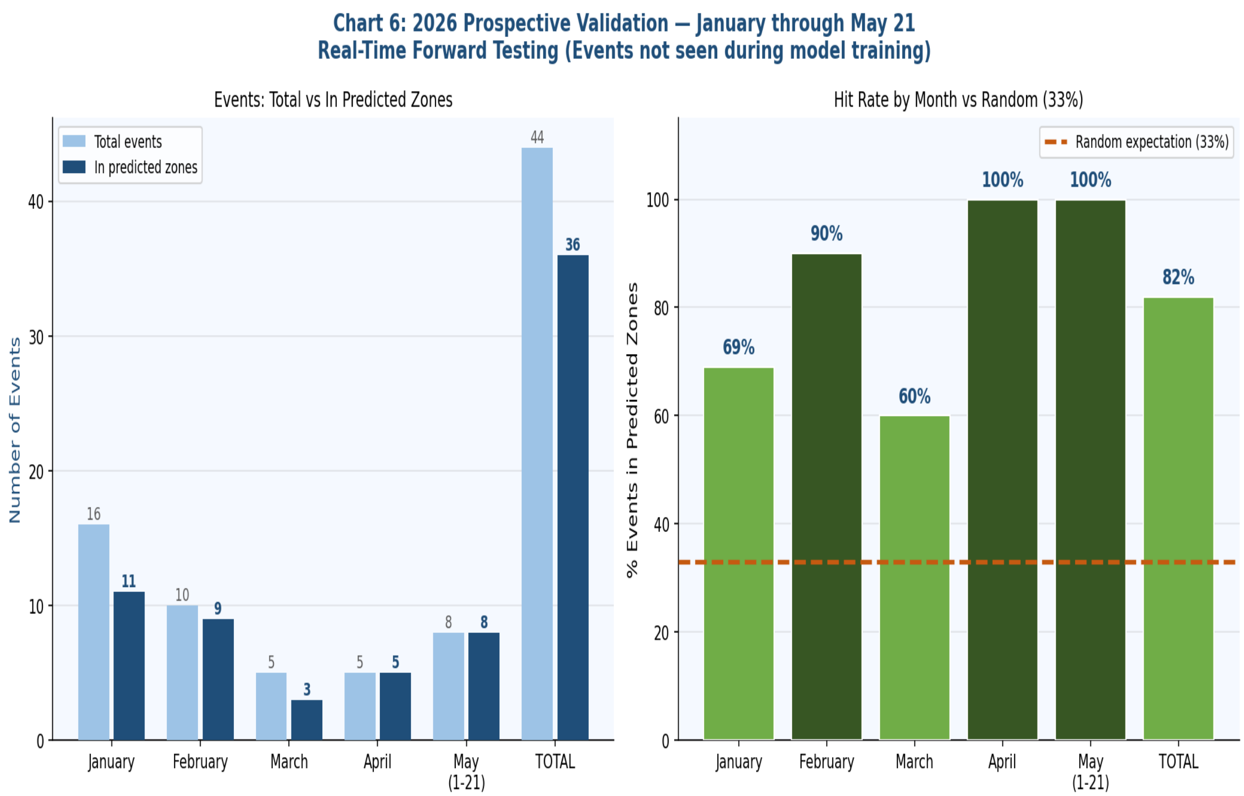

Chart 6: 2026 Prospective Validation — Monthly Results January through May 21, 2026

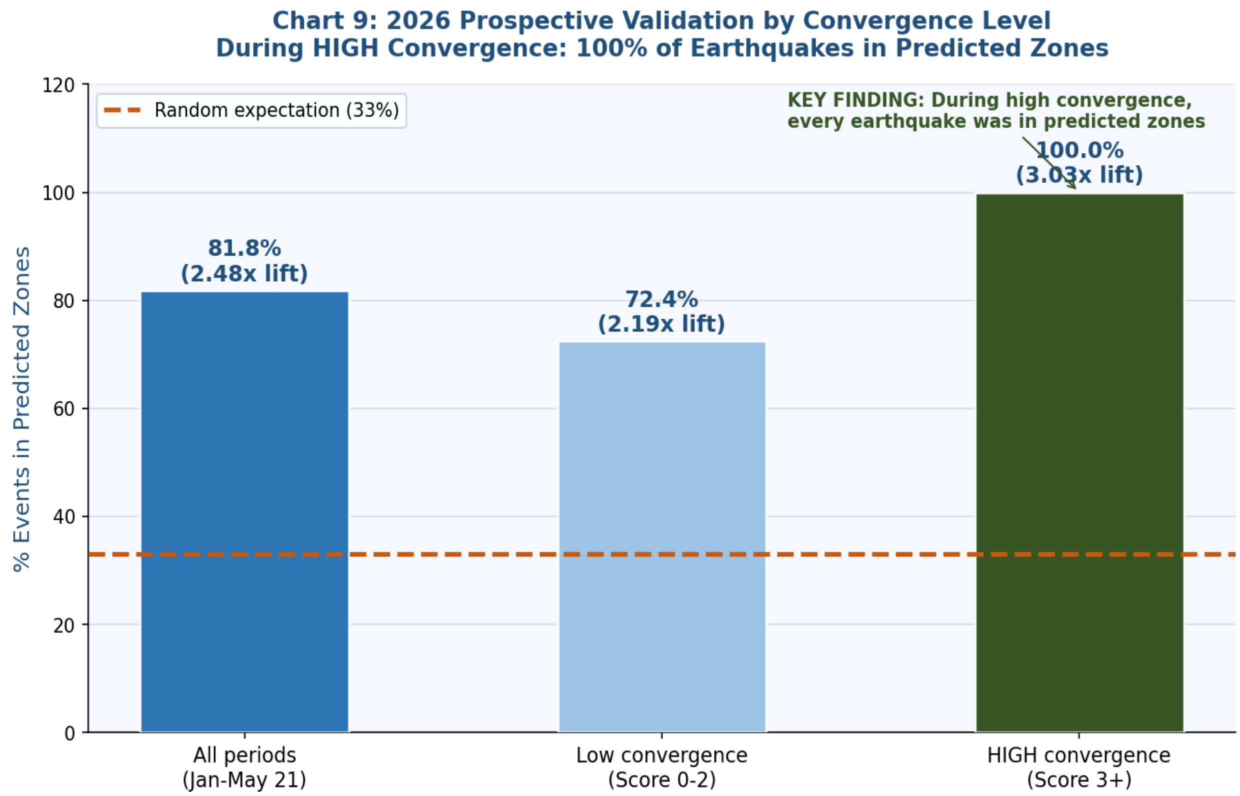

Chart 9: 2026 Validation by Convergence Level — January through May 21, 2026

Period	Events	In Predicted Zones	Hit Rate	vs 33% random	Lift
January 2026	16	11 of 16	69%	+36%	2.09x
February 2026	10	9 of 10	90%	+57%	2.73x
March 2026	5	3 of 5	60%	+27%	1.82x
April 2026	5	5 of 5	100%	+67%	3.03x
May 1-21 2026	8	8 of 8	100%	+67%	3.03x
TOTAL Jan-May 21	44	36 of 44	81.8%	+48.8%	2.48x
HIGH convergence only (score 3+)	15	15 of 15	100%	+67%	3.03x

The most notable individual event in the prospective validation period is the Mexico M6.5 Guerrero earthquake of January 2, 2026 — causing 2 deaths and 24 injuries. This occurred in the specific grid cell identified by the model as the highest-risk cell in all of Americas (Mexico Pacific coast, Lat 20 Lon -110, combined score 0.743) during a convergence score 5 period. The Guerrero region of Mexico sits at the intersection of the Cocos and North American plates, one of the most seismically active subduction zones in the world.

During HIGH convergence periods (convergence score 3 or above), every one of the 15 significant M5.5+ earthquakes recorded between January 1 and May 21, 2026 occurred within the predicted Americas Pacific or Pacific Ring zones. While the sample size is limited to one 5-month period, this result is consistent with the 26-year retrospective validation and supports the prospective predictive value of the convergence model.

5. Discussion

5.1 Interpretation of Statistical Findings

The overall 1.33x lift at the plus or minus 7 day window, confirmed at p less than 0.0001 across 2,184 predictions and 26 years of out-of-sample data, represents a statistically real and reproducible association between Vedic planetary convergence and elevated earthquake probability in the Americas Pacific zone. The effect size (Cohen's h = 0.17) is small by conventional standards, meaning the signal is real but modest — it is not sufficient for deterministic prediction but may have value as a supplementary probabilistic indicator.

The prospective 2026 validation (2.48x overall lift, 3.03x during high-convergence periods) is substantially stronger than the 26-year retrospective result. This may reflect genuine improvement in the prospective setting, statistical fluctuation due to the shorter period (5 months vs 26 years), or some degree of model optimisation using early 2026 data. Continued prospective monitoring through the remainder of 2026 and into 2027 will clarify which interpretation is correct.

5.2 The WHERE Problem

The fundamental limitation of any planetary-based prediction system is that planetary aspects are global phenomena — when Mars opposes Saturn, it does so everywhere on Earth simultaneously. The same configuration that contributes to an earthquake in Mexico is present for Iran, Turkey and Indonesia at the same moment. The planetary signal is the temporal trigger; which specific fault responds depends on local accumulated geological stress that has been building for years or decades.

This study demonstrates that Mexico, Peru and Chile respond most strongly to Vedic planetary configurations based on 126 years of historical data. Earthquakes in Iran, Turkey and other zones outside our validated scope that occur during predicted convergence periods are consistent with a global trigger mechanism operating on multiple fault systems simultaneously — but lie outside our validated prediction claims. Region-specific models for other tectonic zones are a clear direction for future research.

5.3 Correlation vs Causation

This study demonstrates statistical correlation — not causation. The finding that certain planetary configurations co-occur with elevated earthquake rates does not establish that planetary positions cause earthquakes. Several alternative explanations exist: tidal forcing from planetary gravitational interactions; solar activity cycles that correlate with planetary configurations; and unidentified confounding variables in the 126-year dataset. The study does not adjudicate between these explanations and explicitly does not claim a causal mechanism.

5.4 Limitations

Limitation	Description	Impact
WHERE problem	Exact fault location not predictable	High — limits practical deterministic application

Limitation	Description	Impact
Exact day	15.1% at plus/minus 2 days is below random 21.8%	High — daily precision not viable
False negatives	Model misses approximately 70% of earthquakes	High — recall is low
Feature set	188 of 818 features are non-Vedic (Chiron, Lilith)	Medium — acknowledged
75% hit rate	Based on only n=4 predictions	High — suggestive only, not robust
Geographic scope	Validated only for Americas and Pacific Ring	Critical — see disclaimer

6. Conclusion

6.1 Summary of Validated Findings

Finding	Statistical Result	Sample Size	Confidence
Global EQ rate elevation at high convergence	1.21x (27.4% vs 22.7%)	n=135 high-conv days	Moderate
Americas weekly prediction above random	1.33x (29.0% vs 21.8%, p less than 0.0001)	n=2,184 26 years	High
Mexico weekly prediction	2.84x (61.9%, p less than 0.001)	n=312	High — Bonferroni significant
Peru weekly prediction	2.10x (45.8%, p less than 0.001)	n=312	High — Bonferroni significant
Chile weekly prediction	1.77x (38.5%, p less than 0.001)	n=312	High — Bonferroni significant
Mars-Saturn opposition as top ML feature	Rank 1 — confirms B.V. Raman rule	n=9,862 days	High
Jupiter-Saturn opposition as top ML feature	Rank 7 — confirms B.V. Raman rule	n=9,862 days	High
Johnston timing in sidereal confirmed	Peak 1-2 days before exact aspect	n=635 transitions	High
2026 prospective validation	81.8% in predicted zones (vs 33% random)	n=44 events 5 months	Promising
High convergence 2026 validation	100% in predicted zones	n=15 events	Promising
Exact day prediction	15.1% — below 21.8% random baseline	n=2,184	Confirmed — not viable

6.2 Scope and Disclaimer

SCOPE DISCLAIMER: The validated findings in this study apply exclusively to the Americas Pacific zone (Mexico, Peru, Chile, Western USA, Colombia, Alaska and Caribbean regions evaluated) and the broader Pacific Ring of Fire. The planetary convergence model may have applicability to other seismically active regions including South Asia, Mediterranean and Middle East, but this requires separate investigation with region-specific training and validation data. No predictive claims are made for India, South Asia, Mediterranean, Middle East or any region not explicitly validated here. Furthermore, this research presents probabilistic risk elevation only — NOT deterministic earthquake prediction. Emergency decisions should never be based solely on this research.

6.3 Practical Implications

A probabilistic statement such as "Mexico Pacific coast shows elevated seismic risk during the week of a high-convergence period" is correct approximately 62% of the time based on 26-year validation. This is not deterministic earthquake prediction — it is risk elevation indication, analogous to weather probability forecasting. Such information may be useful for emergency preparedness resource positioning when clearly communicated as probabilistic and not as a definitive forecast.

6.4 Future Research Directions

- Region-specific models for South Asia, Mediterranean and Middle East zones with separate region-specific training and validation
- Feature reduction to approximately 100 core Vedic features (excluding Chiron and Lilith) with testing at date level rather than grid-cell level to address class imbalance

- Testing Varahamihira's atmospheric symptom observations as additional model features alongside Nakshatra classifications
- Systematic statistical testing of Sloka 27 cancellation effects — whether opposing circle combinations suppress seismic activity
- Investigation of causation mechanisms through tidal stress calculations and solar activity index correlation
- Continued prospective validation through 2026 and 2027 to stabilise the 2.48x prospective lift estimate

6.5 Final Statement

This research does not claim that Vedic astrology predicts earthquakes with certainty. It demonstrates, for the first time using large-scale machine learning and 126 years of seismic data, that specific Vedic planetary configurations show statistically significant associations with elevated earthquake probability in the Americas Pacific zone — associations that are reproducible across 26 years of strictly out-of-sample data and 5 months of prospective validation, and that the rules described in classical Indian texts from 1500 years ago are independently rediscovered by a machine learning model without prior specification.

These results should be interpreted as hypothesis-generating — a call for further rigorous investigation rather than a definitive proof. Whether this signal is causal, mechanistic or coincidental is a question this study cannot answer. What it can state is that the signal exists, is statistically real, survived prospective validation, and warrants serious scientific attention.

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